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Module 1:
Electrical quantities

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1 Motivation

Electrical engineering permeates virtually every aspect of modern life. Its scope ranges from energy supply and control engineering to information technology. Whether smartphones, cars, or modern household appliances—every day we rely on countless devices that would be inconceivable without this discipline. Over the past decades, electrical engineering has enabled groundbreaking innovations and remains the driving force behind numerous technological developments. In the field of renewable energy, photovoltaic systems, wind turbines, and smart grids are based on advanced electrical engineering principles. Within the Internet of Things (IoT), the networking of everyday objects relies on miniaturised sensors, microcontrollers, and communication modules. Likewise, the development of powerful AI systems would not be possible without specialised hardware components such as GPUs and CPUs. Expertise in this field is not only essential for electrical engineering students. The relevance of its fundamental concepts is steadily increasing for students across all engineering and related disciplines. Modern product development is driven by interdisciplinary project teams.

- Mechanical engineering: Even if mechanical engineers are not directly involved in the development of electrical/electronic components, they need a basic understanding of the colleagues involved in the development project. For example, in the development of robot systems or automated production facilities, close cooperation between mechanical engineering and electrical engineering is essential.
- Computer science: The interface between hardware and software is becoming increasingly important. Computer scientists must understand how their programmes interact with the underlying electronics.
- Medical technology: Modern medical devices such as MRI scanners and pacemakers would be inconceivable without in-depth electrical engineering knowledge.

This script supports students in learning the necessary fundamentals. It not only imparts theoretical knowledge, but also demonstrates practical applications and interdisciplinary connections. The aim is to create a solid foundation on which students from various engineering disciplines can build. The concepts learned here form the basis for understanding more complex systems and prepare students for the challenges of an increasingly digitalised and networked world. From the basics of electricity to advanced topics such as signal processing and control engineering, this script lays the foundation for a successful career in a world that is significantly shaped by electrical engineering.

2 Physical quantities - definitions and units of measurement



Key point: Description of physical quantities

The Students can

- specify physical quantities with measurement values and units of measurement.
- use the seven SI base units and the units derived from them, and assign them to their respective quantities
- Express numerical values using powers of ten and their names, and convert them into each other. Convert

2.1 The SI system of units

Every physical quantity is always described as a combination of a numerical value (measurement) and its unit of measurement (also called dimension). A few unitless quantities are an exception here; the unit of measurement for these is '1'. If necessary, the unit can be preceded by the abbreviation of a power of ten, which scales the numerical value (see section *Zehnerpotenzen*). The numerical value and the unit are separated by a space. Example: 1000 m = 1 km.

These units are defined in the international system of units, the SI system (Système International), and form the basis of all physical quantities. There are seven SI base quantities, which can be defined with the aid of fundamental physical constants. These constants can be determined experimentally with sufficient accuracy to form a reliable basis for our entire system of units. In addition to the base units defined in this way, there are other, so-called derived units, which can be traced back to the base units.

Apart from the SI system and its derived units, there are a number of other units, such as the Anglo-Saxon units pound, inch or mile, which are not used here.

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s
Mass	m	Kilogram, kg	h, s, m
Current value	I	Current, A	e, s
Temperature	T	Kelvin, K	k_B, s, m, kg
Amount of substance	n	Mol, mol	N_A
Light intensity	I_v	Candela, cd	K_{cd}, s, m, kg

Table 2.1: SI units and their base quantities

The currently valid definitions of the units listed in the table above are:

- A second is 9,192,631,770 times the period duration of the hyperfine transition $\Delta\nu$ in the caesium atom. ^{133}Cs .
- One metre is the length of the distance travelled by light in a vacuum during the time $t = \frac{1}{299,792,458}$ s.

- Since May 2019, the kilogram has been dependent on the precisely defined Planck constant:
 $1\text{ kg} = \frac{h}{6.62607015 \cdot 10^{-34} \frac{\text{s}}{\text{m}^2}}$
- Since 2019, the ampere has been defined by the elementary charge. One ampere is the current flow of $\frac{e}{1.602176634} \cdot 10^{-19} \frac{1}{\text{s}}$ elementary charges per second.
- One Kelvin corresponds to a change in thermodynamic temperature that is accompanied by a change in thermal energy (kT) of $1.380649 \cdot 10^{-23} \text{ J}$.
- A mole is the amount of substance in a system that contains $6.02214076 \cdot 10^{23}$ of a specific individual particle.
- A candela is the luminous intensity of a radiation source that emits at a frequency of 540 THz and has a radiation intensity in this direction of $\frac{1}{683} \frac{\text{W}}{\text{sr}}$.

The natural constants used to define the SI system are:

- Hyperfine transition: $^{133}\text{Cs}\Delta\nu = 9.192.631.770 \text{ Hz}$
- Speed of light: $c = 299.792.458 \frac{\text{m}}{\text{s}}$
- Planck's constant: $h = 6,62607015 \cdot 10^{-34} \text{ Js}$
- Elementary charge: $e = 1,602176634 \cdot 10^{-19} \text{ As}$
- Boltzmann constant: $k_B = 1,38064910^{-23} \frac{\text{kg}\cdot\text{m}^2}{\text{s}^2\cdot\text{K}}$
- Avogadro's number: $N_A = 6,02214076 \cdot 10^{23} \frac{1}{\text{mol}}$
- Photometric radiation equivalent of monochromatic radiation of frequency 540 THz: $K_{\text{cd}}(540 \text{ THz}) = 683 \frac{\text{lm}}{\text{W}}$

Derived SI units

From these seven basic units, a further 22 units with their own designations are derived. Some examples that are important for electrical engineering are listed below.:

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$
Resistance	R	Ohm, Ω	$1 \Omega = 1 \text{ V/A} = 1 \text{ kg m}^2/\text{s}^3\text{A}^2$
Capacity	C	Farad, F	$1 \text{ F} = 1 \text{ As/V} = 1 \text{ s}^4\text{A}^2/\text{kg m}^2$
Inductance	L	Henry, H	$1 \text{ H} = 1 \text{ Vs/A} = 1 \text{ kg m}^2/\text{s}^2\text{A}^2$
Magn. Flux	Φ	Weber, Wb	$1 \text{ Wb} = 1 \text{ Vs} = 1 \text{ kg m}^2/\text{s}^2\text{A}$
Flux density	B	Tesla, T	$1 \text{ T} = 1 \text{ Vs/m}^2 = 1 \text{ kg/s}^2\text{A}$

Table 2.2: Sizes and their base units

Important: In equations involving physical quantities, both numerical values and units of measurement appear on both sides of the equal sign. Checking that the units are equal (if in doubt, broken down to the SI base units) is often a good way of checking the plausibility of the calculation.

2.2 Powers of ten

For better readability of very small or large numerical values, units can be scaled by powers of ten. To do this, the name of the power of ten is written directly in front of the unit without spaces. Often, the power of ten is chosen so that the number before the decimal point has as few digits as possible.

Another way to improve readability is to use exponential notation. This involves multiplying the measured value directly by the corresponding power of ten. The number of decimal places used must make physical sense. If nothing else is known, two decimal places are usually used.

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G
Piko, p	10^{-12}	10^{12}	Tera, T
Femto, f	10^{-15}	10^{15}	Peta, P
Atto, a	10^{-18}	10^{18}	Exa, E

Please note: When specifying weights, prefixes are not applied to the SI base unit kilogram (kg), but to the unit gram (g).

For units with exponents, such as area or volume units, the scaling prefix always refers to the base unit, so it must also be raised to the power.

$$1 \text{ m}^1 = 100^1 \cdot 10^{-2} \text{ m} = 100 \text{ cm}$$

$$1 \text{ m}^2 = 100^2 \cdot (10^{-2} \text{ m})^2 = 10.000 \text{ cm}^2$$

$$1 \text{ m}^3 = 100^3 \cdot (10^{-2} \text{ m})^3 = 1.000.000 \text{ cm}^3$$

3 The electrical charge



Learning objectives: The electrical charge

The students can

- describe the properties of electric charges and related physical phenomena
- describe electric fields and calculate them for simple charge configurations
- Calculate forces on charges using Coulomb's law

3.1 The atomic structure

Many physical processes cannot be explained solely by classical phenomena from mechanics. As early as around 600 BC, the Greek philosopher Thales of Miletus discovered that amber, when rubbed with a fur, attracts light objects such as feathers. In further experiments, it was proven much later that, in addition to the force of gravity, there is another force, known as the **electric force** exists in addition to the force of gravity. This force can have both an attractive and a repulsive effect. **Electric charges** are postulated as the cause of these forces and are defined (arbitrarily, from a

historical perspective) as **positive** and **negative**. Charges of the same name repel each other, while charges of different names attract each other.

In 1913, Nils Bohr developed a simple model based on the observed effects. According to this atomic model, shown in Figure 3.1, each atom consists of an atomic nucleus and an electron shell. The atomic nucleus, in turn, consists of densely packed protons (positive charge carriers) and neutrons (electrically neutral). The shell consists of electrons (negative charge carriers) that orbit the atomic nucleus in concentric paths with different radii.

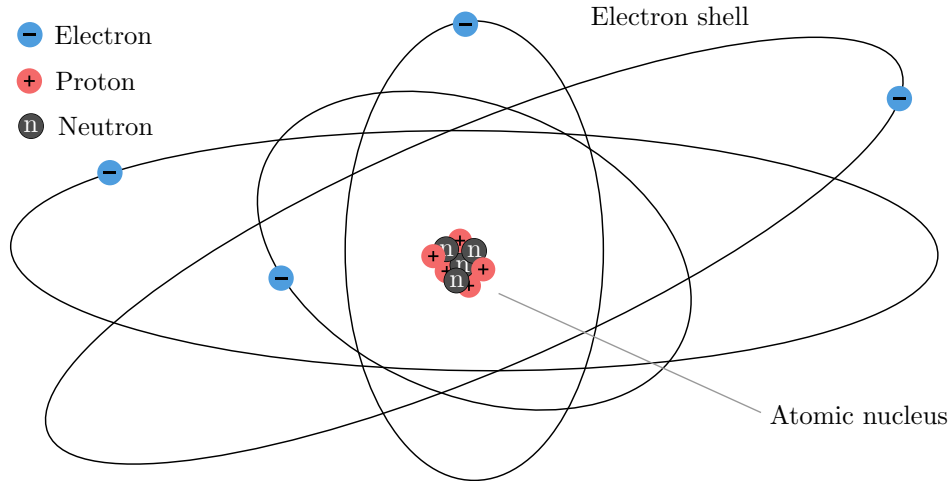


Figure 3.1: Illustration of Bohr's atomic model

The **Load** Q of a proton and an electron is equal in magnitude and is referred to as the elementary charge e . Its experimentally determined value is $e = 1.6 \cdot 10^{-19} \text{ C}$. The proton has a positive charge and the electron has a negative charge. An atom usually has the same number of protons as electrons and is therefore electrically neutral when viewed from the outside.

The coulomb is the unit of electric charge and is described as follows:

$$[Q] = 1 \text{ Coulomb} = 1 \text{ C} = 1 \text{ As}$$

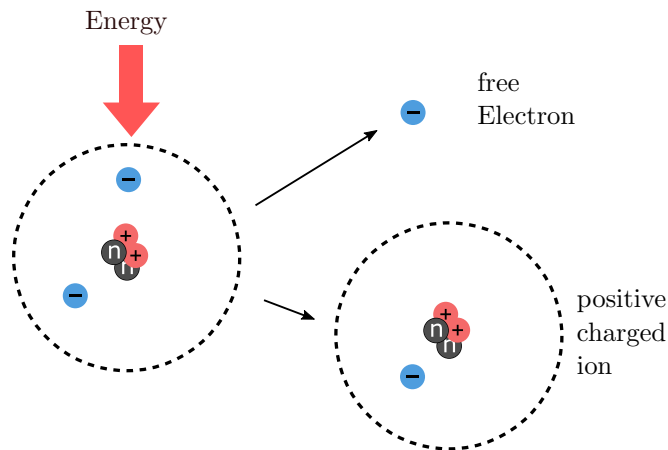


Figure 3.2: Splitting an atom into a positively charged ion and a free electron by supplying energy

The number of electrons in an atom can change through the supply of energy or through interaction with other particles (for example, through friction between a plastic rod and a piece of fabric, friction between clothing and a plastic slide, or UV radiation).

If an atom gains one or more additional electrons, i.e. if there is an **Elektronenüberschuss**, it becomes negatively charged. If it loses electrons, i.e. if there is a **Elektronenmangel**, it becomes positively charged. In both cases, it is referred to as an ion.

Since only whole electrons can be added to or removed from the ion, the corresponding charge Q of an ion (and thus also of any other object) is always a multiple of the elementary charge e .

Key point:

$$Q = \pm n \cdot e$$

The mass of the protons and neutrons in the nucleus is many orders of magnitude higher than that of the electrons. Consequently, the electrons released from the atomic structure are of great importance to electrical engineering as the smallest, lightest and most mobile charge carriers.

3.2 Charge densities

Not only the number, but also the distribution of electrical charge carriers plays an important role in electrical engineering. For example, it is crucial for the design and functioning of electrical components (such as resistors, capacitors or semiconductor devices) to know how the charges are distributed along lines, on surfaces or within volumes. This module assumes a uniform distribution of charge carriers.

The **point charge** serves as an idealised model of a charge distribution at a point without spatial extension. If the actual spatial distribution at the corresponding location is not relevant, charges are usually considered as point charges.

The **line charge density** λ (lambda), given in C/m, describes the distribution of the electric charge along a line, as occurs, for example, in a very thin wire, which is extended in only one dimension (see Figure 3.3).

Since the charge carriers and their atomic structures are negligible compared to the structures used in components, a continuous distribution can be assumed instead of a discrete one. Consequently, the line charge density can be described as the derivative of the charge per line length.

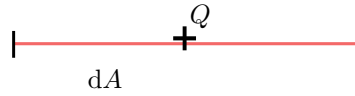


Figure 3.3: Line charge density: A charge Q uniformly distributed along a one-dimensional path with length dl

$$\lambda = \lim_{l \rightarrow 0} \frac{\Delta Q}{\Delta l} = \frac{dQ}{dl}$$

Similarly, the charge Q is the integral of the line charge density λ over the length l .

$$Q = \int_l \lambda dl$$

The **Surface charge density** σ (sigma) in C/m^2 describes the distribution of the charge Q , which is distributed per unit area A (see Figure 3.4). It is required, for example, when calculating capacitors, which will be introduced at a later stage.

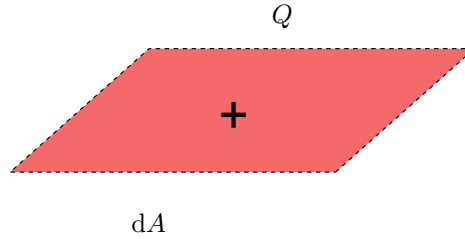


Figure 3.4: Surface charge density: A charge Q evenly distributed over a two-dimensional surface with the size dA .

Similar to the line charge density, σ can also be described using a limit analysis and thus a derivative:

$$\sigma = \lim_{A \rightarrow 0} \frac{\Delta Q}{\Delta A} = \frac{dQ}{dA}$$

$$Q = \iint_A \sigma dA$$

The **Space charge density** ρ (rho) in C/m^3 indicates the number of free charge carriers in the volume under consideration. These can be, for example, electrons in a conductor or charged ions in a gas mixture.

$$\rho = \lim_{V \rightarrow \infty} \frac{\Delta Q}{\Delta V} = \frac{dQ}{dV}$$

$$Q = \iiint_V \rho dV$$

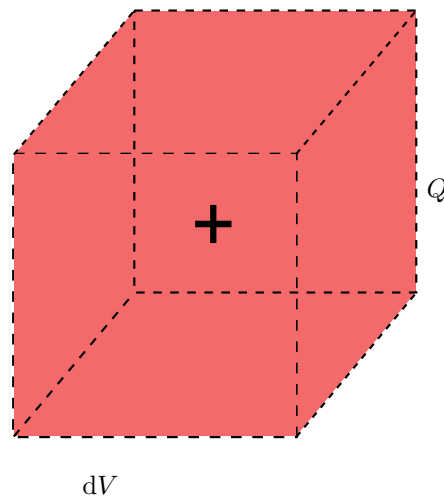


Figure 3.5: Space charge density ρ : A charge Q uniformly distributed in a three-dimensional volume with the size dV .

3.3 Electrical conductors - metals

This distribution of charge carriers can be clearly seen in metals, for example. Many metals have a special atomic structure. The atoms are arranged in a regular **Lattice structure**. As a result, the outermost electrons of each atom within the metal body are almost free to move. These free electrons give metals their electrically conductive properties. Figure 3.6 illustrates this structure. Metals are therefore often referred to in electrical engineering as (electrical) conductors. In general, conductors are electrically neutral, since in each atom the sum of all electrons corresponds to the sum of all protons.

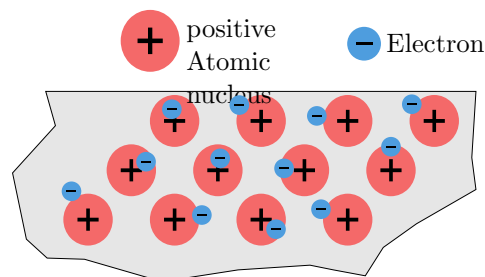


Figure 3.6: Cut-out from a sheet of metal. The positively charged atomic nuclei and their freely moving outer electrons are shown.

The addition of electrons creates an excess of electrons, causing the conductor to become negatively charged. If electrons are removed, an electron deficiency occurs and the conductor becomes positively charged. (Abbildung 3.7).

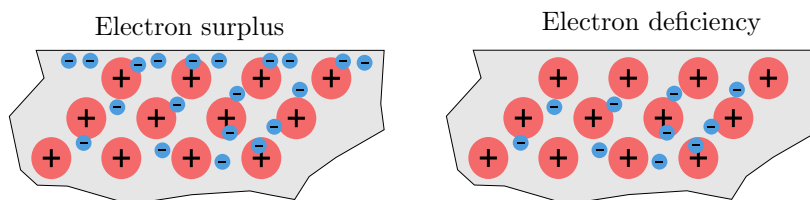


Figure 3.7: Conductors with an excess of electrons (left) or a deficiency of electrons (right) on their surface.

3.4 Coulomb's law

Charged particles influence each other, with charges of the same sign repelling each other and charges of different signs attracting each other. As early as 1785, Charles Augustin de Coulomb was able to experimentally prove that the magnitude of the force is proportional to each of the spherically symmetric charges Q_1 and Q_2 , but inversely proportional to the square of the distance r between the two charges.

The proportional interaction between the charges Q_1 and Q_2 and the square of the distance r^2 leads to the force between the charges with the proportionality constant $\frac{1}{4\pi\epsilon_0}$. This proportional relationship is shown in Figure 3.8.

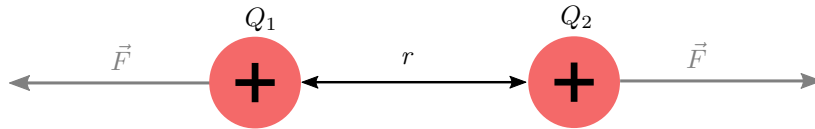


Figure 3.8: Arrangement of two point charges Q_1 and Q_2 and the forces acting on them.

By introducing a proportionality term, the resulting force F can be calculated directly from the previously determined proportional relationship. The resulting equation 3.1 is called Coulomb's law.

$$F \sim \frac{Q_1 Q_2}{r^2} \rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} \quad (3.1)$$

The factor ϵ_0 is referred to as the **electric field constant** (also known as the dielectric constant of vacuum), while the 4π originates from geometric considerations of the arrangement.

The value of the electric field constant is approximately:

$$\epsilon_0 = 8,854 \cdot 10^{-12} \text{ As/Vm}$$

If there are more than two charges, the resulting force on each of the individual charges can be calculated by adding all the individual forces resulting from each combination of two charges in which the target charge is involved. This effect is known as the superposition principle, which is also used in Module 5 *Advanced DC Networks* to calculate resulting voltages.

In practice, this effect is used, among other things, in the electrostatic tensioning of paper on plotters, in laser printers to transfer toner powder to the paper, in touch screens to determine the point of contact, and in plasma screens for the controlled gas discharge of ions to generate light.

4 The electric field

Learning objectives: The electric field

The students can

- develop a ‘feel’ for electric fields
- describe electric fields and calculate them for simple charge configurations
- characterise the behaviour of electric fields on conductors

4.1 Characterisation of the electric field

As shown, electric charges have a direct influence on other charges. This occurs because they alter the properties of the space around them. These changes can be described by a field model. The concept of a field is of fundamental importance in physics. Examples of this are the temperature field, in which the temperature is described by a scalar quantity depending on its location.

In addition to such scalar fields, there are also vector fields in which each location is assigned not only a quantity but also a direction in which it acts. Examples of this are the gravitational field or a flow field, which describes, for example, the direction and velocity of water particles.

The **electric field** is described by **magnitude** (in newtons) and the **direction** of the **force** $\vec{F}_E$ on a positive test charge Q_2 and characterised by field lines. This arrangement is shown in Figure 4.1.

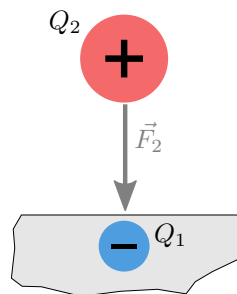


Figure 4.1: Attractive force F_2 of a test charge Q_2 in the direction of a stationary charge Q_1

The electric field strength is therefore:

Key point:

Electric field strength = $\frac{\text{Force on load}}{\text{Load}}$

$$\vec{E}_1 = \frac{\vec{F}_2}{Q_2} \quad \left[\frac{\text{N}}{\text{C}} = \frac{\frac{\text{kg} \cdot \text{m}}{\text{s}^2}}{\text{A} \cdot \text{s}} = \frac{\text{V}}{\text{m}} \right] \quad (4.1)$$

At first glance, a disadvantage of this definition appears to be that the electric field strength seems to depend on the test charge Q_2 . However, by inserting Coulomb's law (equation 3.1) into equation 4.1, it can be shown that the force F_2 always increases proportionally to the test charge Q_2 . Their ratio therefore always yields the same value for the electric field strength, which, as shown in Figure 4.2, becomes independent of the test charge:

$$\vec{E}_1 = \frac{\vec{F}_2}{Q_2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 \cdot Q_2}{r^2} \cdot \frac{1}{Q_2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1}{r^2} \quad (4.2)$$

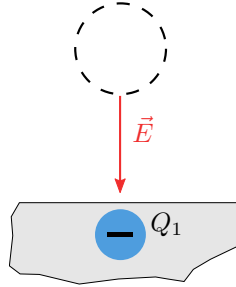


Figure 4.2: The electric field caused by Q_1 is independent of any sample charge.

4.2 Electrical conductors and electrostatic fields I

The previous chapter examines the effect of point charges on the electric field. However, since electric fields in everyday life often originate from charged electrical conductors (such as metals), they are listed separately here. If there is an excess of electrons, the positive test charge Q is attracted towards the plate surfaces, as shown in Figure 3.7. The field lines $\vec{E}$ of the electric field also run in the direction of the conductor surface. In the case of an electron deficiency, the test charge is repelled and the field lines also lead away from the positive conductor.

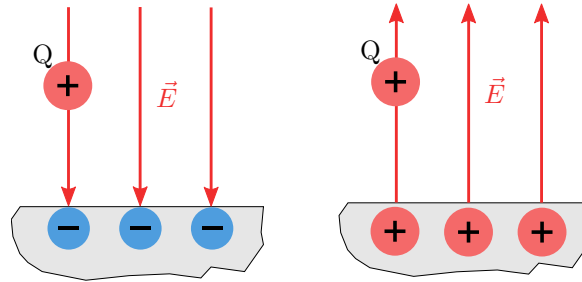


Figure 4.3: Direction of the electric field strength $\vec{E}$ from a negatively (left) and positively (right) charged conductor to a positive test charge

Since electrons are freely mobile within a conductor, in the case of an electric field within a conductor, they would be attracted (or repelled) by this field until the field within the conductor has balanced out (see Figure 4.4). Consequently, in this case, no electric field can exist within a conductor.

This leads to the realisation that the electric field lines must necessarily be **perpendicular** to the surface of conductors (see Figure 4.5). Otherwise, an oblique field line could be vectorially decomposed into a component running perpendicular and a component running horizontally on the conductor surface. Since this horizontal component is instantly balanced by the moving charge carriers, it cannot exist. This effect occurs exclusively in electrically conductive materials; in non-conductive materials (so-called insulators), there are no moving charge carriers that can balance the horizontal component.

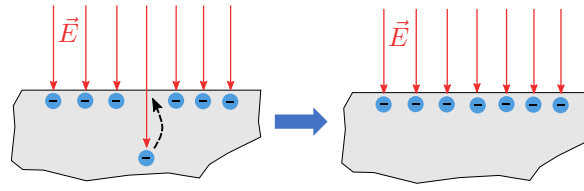


Figure 4.4: Free charge carriers are instantly attracted to the edge of the conductor by the electric field. Therefore, no electrostatic field can exist within the conductor. electrostatic field.

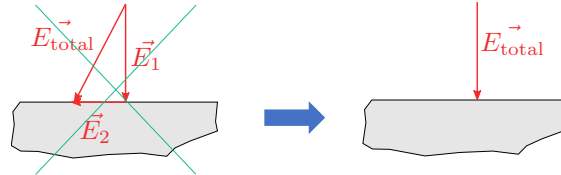


Figure 4.5: By eliminating the horizontal component $\vec{E}_2$ of the electric field, only the vertical component $\vec{E}_1$ remains, which is perpendicular to the conductor surface.

4.3 Examples of electric fields

The simplest electric field is the **homogeneous field**. In this field, both the magnitude and the direction of the electric field strength $\vec{E}$ are constant at every point. A test charge located in this field therefore experiences the same force at every point. Such a field can be generated, as shown in Figure 4.6, for example between two parallel metal plates that are charged with different charges Q . The field lines generally run from the positively charged to the negatively charged metal plate. The density with which the field lines are drawn is often used as a measure of the strength of the electric field. The denser the field lines are drawn, the stronger the electric field is at that point.

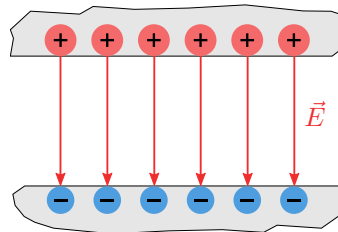


Figure 4.6: An electric homogeneous field between two differently charged metal plates

The electric field of a point charge is also referred to as the **radial field** (see Figure 4.7). Under the idealised assumption that there is a shell with opposite charge infinitely far away, the field lines propagate in a straight line. In their symmetrical arrangement, they point in the direction of the negative shell for positive point charges, and in the direction of the point charge for negative point charges.

The electric field strength $\vec{E}$ is directly proportional to the charge Q , but decreases quadratically with increasing distance r from it. Its magnitude can be calculated using:

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (4.3)$$

The electrical field strength, which weakens with increasing distance r from the point charge, is evident not only in the mathematical relationship but also in the increasing distance between the field lines.

More complicated electric fields arise when a second point charge is added. In the case of two charges

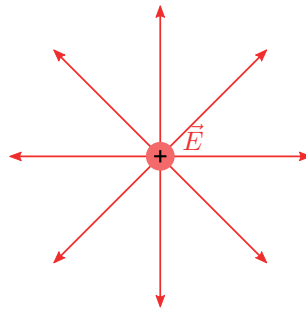


Figure 4.7: Radial electric field of a positive point charge.

of equal magnitude but opposite sign (Figure 4.8), an almost uniform field exists between the charges, while it becomes significantly weaker further out.

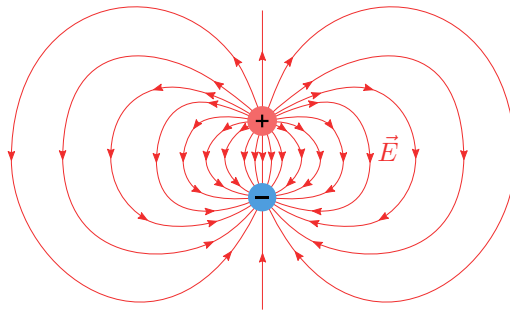


Figure 4.8: The electric field of two charges of equal magnitude but opposite sign

The case with two identical point charges shows a special feature of the electric field, as illustrated in Figure 4.9. While the field lines in the rest of the space point away from the positive charges, as expected, there is a free space between the point charges without any field lines. This is because the opposing field lines in this space between the charges cancel each other out. There is no electric field strength at the point between the two charges; on the plane between the charges, the respective horizontal components neutralise each other, and there is only an electric field in the respective vertical direction.

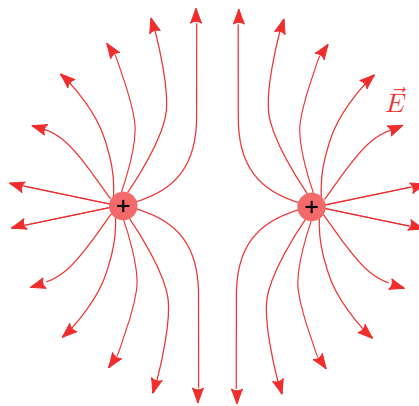


Figure 4.9: The electric field of two charges of equal magnitude and the same sign

5 The electrical potential

Learning objectives: The electrical potential

The Students can

- Calculate displacement work of charges in an electric field
- Determine and draw equipotential surfaces
- determine electrical voltages from given field sizes

5.1 Work in an electric field

If a charge is located within an electric field, a force is exerted on it. If the charge is then moved within the field, work must be done. The gravitational field can be viewed analogously here, in that work must be done when a mass is lifted, i.e. the potential energy of the object is increased.

This section should be seen as a brief preview of Module 2 *Energy and Power*, in which electrical work is defined precisely. The sign of the work done depends on whether the work is done on the system under consideration (here: potential energy of the test charge Q in the electric field $\vec{E}$)

or to the generator of this work. This script refers to the generator. If the generator has to perform work to increase the potential energy of the system, the work done is considered negative. If work is released by the system, i.e. the potential energy of the sample charge decreases, the work is positive.

Physically speaking, the work is the product of the force required to move the charge multiplied by the distance travelled. Using the knowledge from Section 4.1 that the electric field strength is the quotient of the force on a charge per charge (see equation 4.1), the work required to move the charge can be determined.

Below, we will calculate the work W released by the positive test charge Q from Figure 5.1 when it moves along the field lines in the direction of the negatively charged conductor plate.

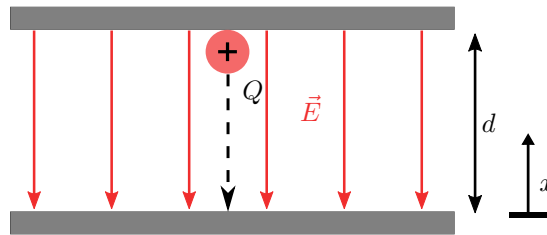
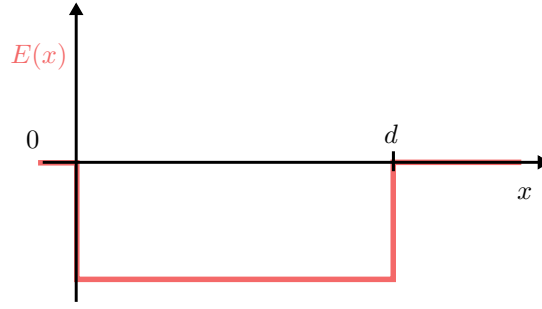


Figure 5.1: Test charge Q in an electric homogeneous field $\vec{E}$

The course of the electric field strength and direction of $\vec{E}$ relative to the path x can be represented as shown in Figure 5.2.

$$W = \int_d^0 \vec{F} d\vec{x}$$

Since the direction of the electric field $\vec{E}$ is opposite to the spatial direction x , eliminating the vector arrows results in a negative sign in front of the integral. Substituting the force F into equation 4.1 yields:

Figure 5.2: Field strength $E(x)$ in the structure shown in Figure 5.1

$$W = -Q \int_d^0 E \cdot dx$$

Solving the integral and inserting the limits of integration yields:

$$W = -Q \cdot E [x]_d^0 = -Q \cdot E(0 - d)$$

$$W = Q \cdot E \cdot d$$

The difference between the work done or removed by a charge in an electric field is therefore:

$$\Delta W = Q \cdot E \cdot \Delta x \quad (5.1)$$

Example 5.1

For the sample charge shown in Figure 5.1, the aim is to determine how much energy is released on its way from the starting point to the negative plate.

The test charge is $Q = 1\mu\text{C}$, the distance to the plate is 0.1 m, and the electric field is $E = 200 \text{ V/m}$.

The work W is then calculated as

$$W = Q \cdot E \cdot d = 1 \cdot 10^{-6} \text{C} \cdot 200 \text{V/m} \cdot 0,1 \text{m} = 2 \cdot 10^{-5} \text{J}$$

Similar to an object in a gravitational field, the electric field of a charge causes energy. This accelerates the charge, and the potential energy is converted into kinetic Energy.

It is immediately apparent that the amount of potential energy stored in the electric field and usable by the test charge Q is linearly dependent on the location x of the test charge (see Figure 5.3). If the test charge already has kinetic energy in advance, or if the electric field extends beyond the section under consideration, the equation can be expanded by the additional term W_0 :

$$W(x) = Q \cdot E \cdot x + W_0 \quad (5.2)$$

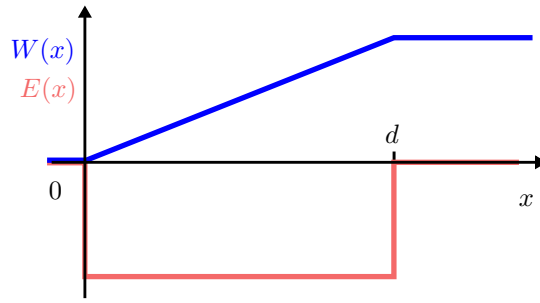


Figure 5.3: Linear relationship between the position x and the potential energy $W(x)$ of the test charge.

The transition from the potential energy stored in the electric field to the kinetic energy of the sample charge can be illustrated using the diagram in Figure 5.4.

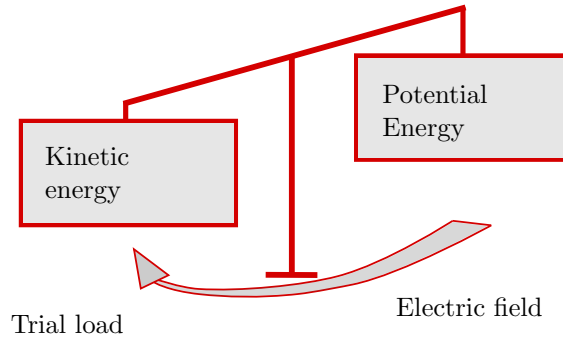


Figure 5.4: Transition from potential energy in the electric field to potential energy in the test charge

For general description purposes, the technically usable field energy must be independent of the sample charge Q . To this end, the potential energy is normalised to a working potential.

$$\frac{W(x)}{Q} = E \cdot x + \frac{W_0}{Q}$$

5.2 Definition of electrical potential

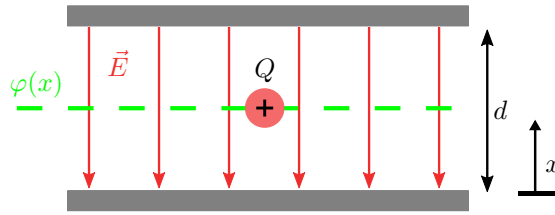
The potential energy normalised to the charge Q under consideration is the **electric potential** φ of the electric field. This can be determined directly for every point in space, independently of other influencing factors. In contrast to the electric field, the potential field is a scalar field. This means that a unique electric potential can be assigned to every point in space independently of other influencing factors, but there is no direction for this quantity in space.

$$\varphi = \frac{W}{Q} \quad (5.3)$$

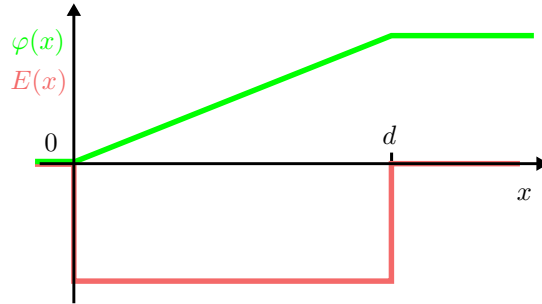
$$[\varphi] = \text{V (Volt)}$$

Figure 5.5 shows an electric homogeneous field in which the potential line on which the charge Q is currently located is plotted. As the distance from the positive plate increases in the x direction, the potential rises and can be calculated as follows:

$$\varphi(x) = E \cdot x + \varphi_0 \quad (5.4)$$

Figure 5.5: Electric homogeneous field with potential line plotted $\varphi(x)$

The gradient of the potential is determined by the electric field strength $\vec{E}$. The electric potential φ and the electric field strength are related by the negative gradient of the potential. This relationship is illustrated in Figure 5.6.

Figure 5.6: Relationship between the electric potential φ and the electric field strength E in a homogeneous field

The relationship can be expressed in a homogeneous field by the following equation:

$$E = -\frac{\Delta\varphi}{\Delta x} \quad (5.5)$$

In inhomogeneous, three-dimensional fields, the general equation applies, in which the gradient of φ is calculated in every spatial direction:

$$E = -\text{grad } \varphi(x,y,z) = \left(\frac{\partial\varphi}{\partial x}, \frac{\partial\varphi}{\partial y}, \frac{\partial\varphi}{\partial z} \right) \quad (5.6)$$

In order to fully describe the electric potential at a point, as shown in equation 5.4, a reference potential φ_0 must be selected as a necessary boundary condition, which is clearly defined. In principle, this potential can be selected arbitrarily, but as a rule, a reference electrode is selected for which the potential is set to zero. In the case of the setup shown in Figure 5.5, this would be:

$$\varphi_0 = \varphi(x=0) = 0$$

5.3 Connection between work and potential

In a homogeneous electric field, the potential φ is, as shown in formula 5.4, exclusively dependent on the distance x to the reference electrode. This means that all surfaces perpendicular to the field have a constant potential. Such surfaces are called equipotential surfaces, as shown in Figure 5.7 by the green lines. No work needs to be done to move a charge along such a surface.

The potential energy W_{12} of a test charge Q moved by the electric field from point x_1 to x_2 can be described by the potential difference between these points. The order of the indices of the work W_{12}

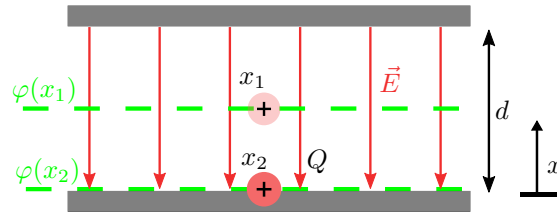


Figure 5.7: Electric homogeneous field in which a test charge Q moves from the equipotential line $\varphi(x_1)$ to the equipotential line $\varphi(x_2)$.

describes the direction of movement.

If a test charge is moved from the equipotential surface $\varphi(x_1)$ to $\varphi(x_2)$ as shown in Figure 5.7, it absorbs the following energy:

$$W_{12} = Q \cdot (\varphi_1 - \varphi_2)$$

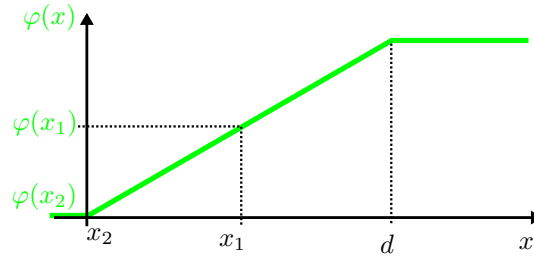


Figure 5.8: Representation of the potential curve $\varphi(x)$ within the electric homogeneous field

With the choice of reference surface $\varphi_2 = 0$ made in Figure 5.8, the result for the work supplied is:

$$W_{12} = Q \cdot (\varphi_1 - 0) = Q \cdot \varphi_1$$

The following thought experiment illustrates another key property of the electric field.

As shown in Figure 5.9, a test charge Q is moved in a closed circuit between points 1-4 in a homogeneous field. The partial energies required for this process are considered. Thus, when moving from point 1 to point 2, work of $W_{12} = Q \cdot (\varphi_1 - \varphi_2)$ is performed, and from point 4 to point 1, energy $W_{41} = Q \cdot (\varphi_4 - \varphi_1)$ is performed.

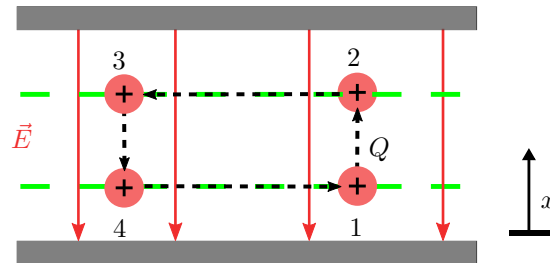


Figure 5.9: Representation of the potential profile $\varphi(x)$ within the electric homogeneous field

The energy required for a complete cycle is calculated by adding up the four partial energies:

$$W_{\text{ges}} = Q \cdot (\varphi_1 - \varphi_2) + Q \cdot (\varphi_2 - \varphi_3) + Q \cdot (\varphi_3 - \varphi_4) + Q \cdot (\varphi_4 - \varphi_1)$$

Since the charge Q is identical in all terms, factoring out Q yields:

$$W_{\text{ges}} = Q \cdot (\varphi_1 - \varphi_2 + \varphi_2 - \varphi_3 + \varphi_3 - \varphi_4 + \varphi_4 - \varphi_1)$$

$$\rightarrow W_{\text{ges}} = Q \cdot 0 = 0$$

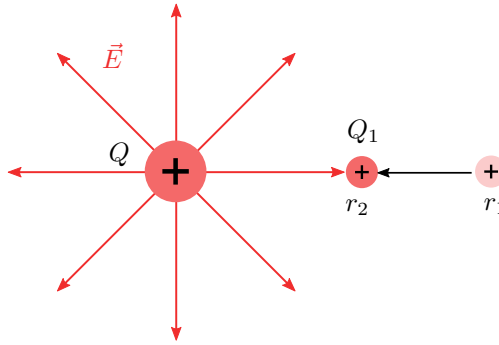
No energy is required for a closed cycle of a test charge in a homogeneous field, as it depends solely on the starting and ending points of the movement, but not on the chosen path.

Consequently, the electric field is also referred to as a **vortex-free source field**. This means that the field lines do not form closed lines, but each have a starting point (on positive charges) and an end point (on negative charges).

This observation applies not only to homogeneous fields, but to all electrostatic fields.

Example 5.2: Work in the radial field

In the following example, the test charge Q_1 is to be moved from a distance r_1 to a distance r_2 to the fixed charge Q .



Q is 1 C, Q_1 is 1 mC. What work is required if $r_1 = 1$ m and $r_2 = 50$ cm?

$$W_{12} = Q_1 \int_{r_1}^{r_2} \vec{E} \cdot d\vec{r}$$

$$W_{12} = Q_1 \int_{r_1}^{r_2} \left(\frac{Q}{4\pi\epsilon_0 r^2} \right) dr$$

$$W_{12} = Q_1 \cdot \frac{Q}{4\pi\epsilon_0} \int_{r_1}^{r_2} \frac{1}{r^2} dr$$

The primitive function of $\frac{1}{r^2}$ is $-\frac{1}{r}$.

$$W_{12} = Q_1 \cdot \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{r} \Big|_{r_1}^{r_2} \right) = Q_1 \cdot \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{r_2} + \frac{1}{r_1} \right)$$

where $\frac{Q}{4\pi\epsilon_0} \frac{1}{r}$ corresponds to the potential $\varphi(r)$.

$$W_{12} = Q_1 \cdot (\varphi(r_1) - \varphi(r_2))$$

Inserting the given values:

$$W_{12} = (1 \cdot 10^{-3} \text{ C}) \cdot \frac{1 \text{ C}}{4\pi\epsilon_0} \left(\frac{1}{1 \text{ m}} - \frac{1}{0,5 \text{ m}} \right)$$

$$W_{12} = -8,98755 \times 10^6 \text{ J}$$

Consequently, approximately 8.99 megajoules must be expended to move the charge Q_1 .

5.4 Definition of electrical voltage

The energy that can be utilised from the electric field for a technical process is given by the **potential difference** between the starting point and end point of the path. Due to its fundamental importance in electrical engineering, it is given a name and formula letter, and is referred to as **electrical voltage** U :

$$U_{12} = \varphi_1 - \varphi_2 \quad (5.7)$$

$$[U] = \text{V (Volt)}$$

The order of the indices indicates the direction of the voltage arrow, analogous to the potential difference. In Figure 5.10, the voltage U_{12} describes the potential difference between the potentials φ_1 and φ_2 . Due to the properties of equipotential surfaces, it is completely irrelevant which point of φ_1 is started from and which point on φ_2 is the destination. The path chosen between the points is also irrelevant for the voltage between the points.

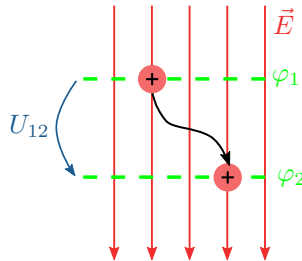


Figure 5.10: Voltage U_{12} between the two equipotential lines φ_1 and φ_2

The relationship between the electric field strength and the potential is already known from equation 5.4 and is given (for a reference potential of $\varphi_0 = 0$) by $\varphi(x) = E \cdot x$.

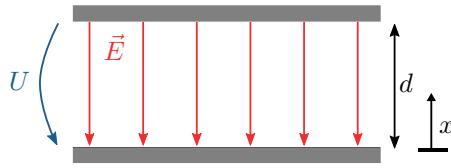
Inserted into the definition of voltage (equation 5.7), this results in:

$$U = \varphi(x = d) - \varphi(x = 0) = E \cdot d - E \cdot 0$$

The electrical voltage U can be calculated in a homogeneous field as shown in Figure 5.11 as follows, provided that the distance d is parallel to the field lines:

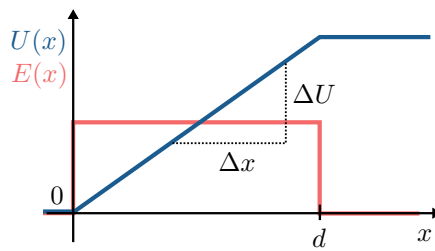
$$U = E \cdot d \quad (5.8)$$

Alternatively, the electric field strength can also be calculated from a given voltage U and a known distance d :

Figure 5.11: Electrical voltage U in a homogeneous field

$$E = \frac{U}{d} \quad (5.9)$$

Figure 5.12 shows that the voltage is nothing more than the integrated field strength E over the distance d .

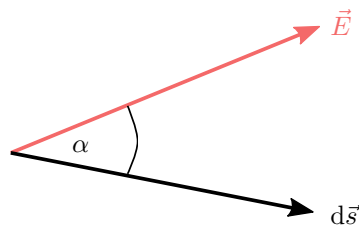
Figure 5.12: Relationship between electrical voltage U and electrical homogeneous field E

While this integral can be determined in a homogeneous field by multiplying the electric field E and the distance d , in inhomogeneous fields this is only possible by solving the integral:

$$U_{12} = \int_1^2 \vec{E} \cdot d\vec{s} \quad (5.10)$$

It should be noted here that the product $\vec{E} \cdot d\vec{s}$ consists of vectors. If they point in the same direction, the vector arrows can be eliminated; otherwise, the scalar product must be used:

$$\vec{E} \cdot d\vec{s} = E \cdot ds \cdot \cos(\alpha)$$

Figure 5.13: Scalar product of field strength $\vec{E}$ and path segment $d\vec{s}$

6 The electric current



Learning objectives: The electrical current

The Students can

- Explain and apply the concepts of electric current and electric current density. Apply
- calculate the electric current and electric current density in simple arrangements
- determine the drift velocity of electrons in simple arrangements

6.1 The electric flow field

The previous chapters refer exclusively to stationary charge carriers. Consequently, this subfield of electrical engineering is referred to as electrostatics.

At the molecular level, the free electrons in an electrical conductor constantly move in random directions. On average, however, the movement of the individual electrons balances out. If an electric field $\vec{E}$ is applied to a conductor, the free moving electrons in it are attracted towards the positively charged electrode (see Figure 6.1). In addition to the random, undirected motion, a **directed motion component** is added. If such a directed motion of particles occurs, it is referred to as a **flow field**.

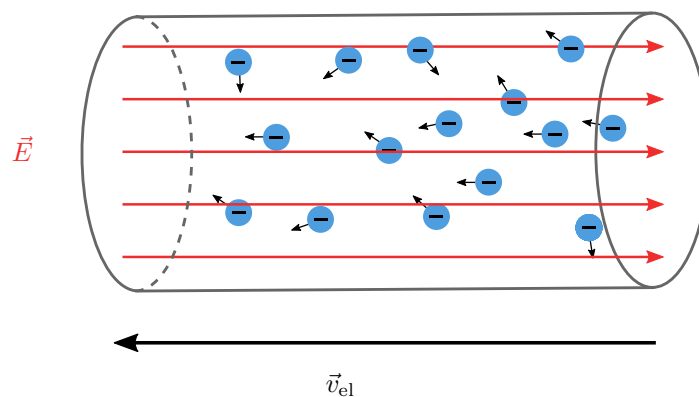


Figure 6.1: Electrical conductor with an impressed electric field strength $\vec{E}$. The drift motion of the electrons $\vec{v}_{el}$ proceeds in the opposite direction to the field strength.

In addition to electrons in conductors, an electric flow field can also be caused by charged ions in gases or liquids. In the case of a charge carrier movement that is constant on average over time, which is achieved in the conductor shown by a constant electric field, a stationary electric flow field is present.

6.2 Electric current

Electrical current I refers to the directed movement (drift movement) of electrical charge carriers. The direction of the electric current is defined as running from the higher potential (positively charged electrode) to the lower potential. It therefore runs in the same direction as the electric field strength, but in the opposite direction to the drift velocity of the electrons (see Figure 6.2).

The electrons reach a field strength-dependent drift velocity $\vec{v}_{el}$. Collisions with the metal atoms from the lattice structure impede the free movement of the electrons and counteract it as a kind of resistance.

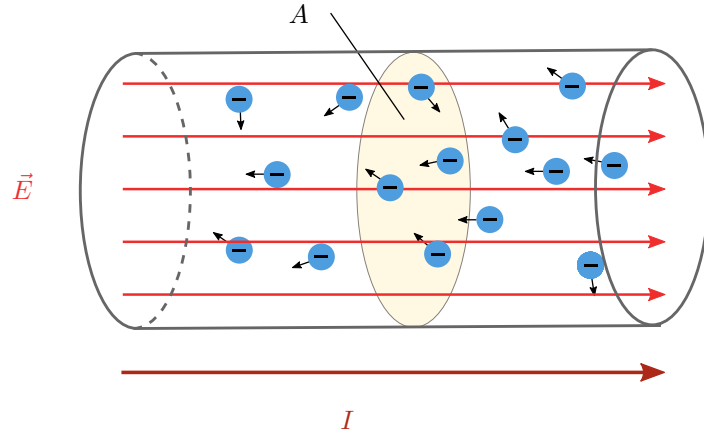


Figure 6.2: Electrical conductor with electric field strength $\vec{E}$ and resulting electric current I .

$$\vec{v}_{\text{el}} = -b_{\text{el}} \cdot \vec{E} \quad (6.1)$$

$$[\vec{v}_{\text{el}}] = \frac{\text{m}}{\text{s}}$$

The proportionality factor b_{el} between the drift velocity $\vec{v}_{\text{el}}$ and the electric field strength $\vec{E}$ is referred to as electron mobility.

$$[b_{\text{el}}] = \frac{\text{m}^2}{\text{Vs}}$$

In addition to the drift velocity $\vec{v}_{\text{el}}$, the number of electrons N_{el} and their charge quantity ΔQ are also relevant for determining the electric current I .

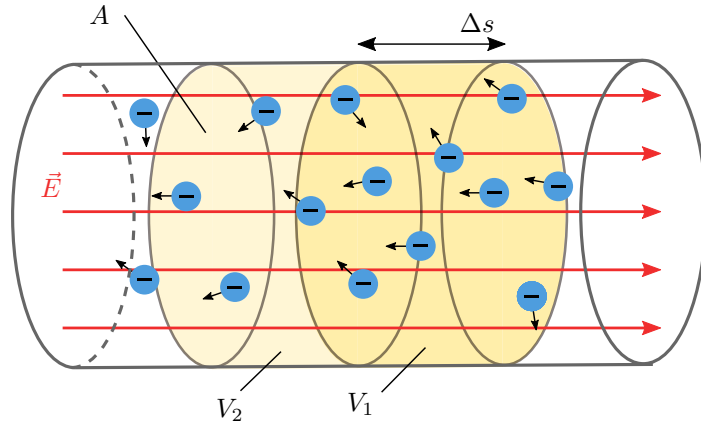


Figure 6.3: Illustration of the movement of electrons through an electrical conductor

The charge quantity ΔQ corresponds to

$$\Delta Q = e \cdot N_{\text{el}} = e \cdot n_{\text{el}} \cdot V_1$$

where n_{el} indicates the density of charge carriers in volume V_1 .

With $V_1 = \Delta s \cdot A$ and $\Delta s = v_{\text{el}} \cdot \Delta t$, we get:

$$\Delta Q = e \cdot n_{\text{el}} \cdot v_{\text{el}} \cdot \Delta t \cdot A$$

Using the definition of drift velocity (6.1), this leads to:

$$\Delta Q = e \cdot n_{\text{el}} \cdot b_{\text{el}} \cdot E \cdot \Delta t \cdot A,$$

from which the change in charge quantity ΔQ per unit of time Δt can be derived:

$$\frac{\Delta Q}{\Delta t} = e \cdot n_{\text{el}} \cdot b_{\text{el}} \cdot E \cdot A$$

6.3 Definition of electrical current

The change in charge over time determined in this way is referred to as **electric current** I :

$$I = \frac{\Delta Q}{\Delta t} \quad (6.2)$$

$$[I] = \text{Ampere} = \text{A}$$

From this definition of current, it follows that the electrical current accumulated over a period of time results in the amount of charge Q transported during that time.:

$$Q = \int_{t_1}^{t_2} i(t) dt \quad (6.3)$$

In addition to the total electric current I , the current relative to the conductor cross-section is often also of interest. To calculate this **current density** J , the current ΔI is considered through an elementary small area element ΔA :

$$J = \frac{\Delta I}{\Delta A} \quad (6.4)$$

If the electric current I is evenly distributed across a surface, this can be simplified:

$$J = \frac{I}{A} \quad (6.5)$$

Since the current I remains identical across the entire conductor, a reduction in the cross-sectional area of the conductor also changes the resulting electric current density J (see Figure 6.4).

The following applies here:

$$J_1 \cdot A_1 = J_2 \cdot A_2 = I$$

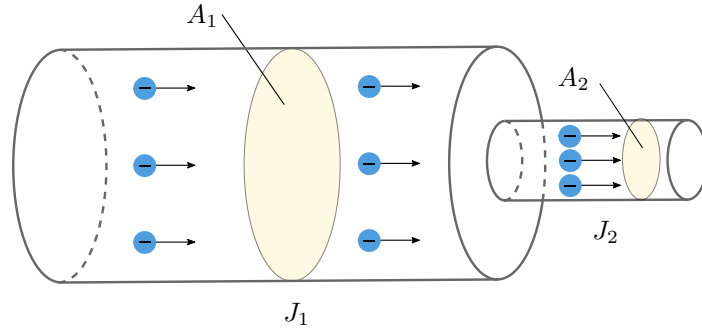
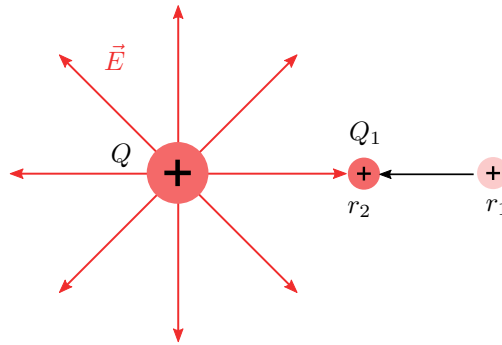


Figure 6.4: Correlation between the current density J_1 at cross-section A_1 and J_2 at A_2

Example 6.1

A current of $I = 8\text{A}$ flows through a copper wire with a cross-sectional area of $A = 1\text{mm}^2$. One mm^3 contains approximately $8.5 \cdot 10^{19}$ atoms. Let us assume that 1 electron per atom is involved in charge transport.



What is the average drift velocity of the electrons through the wire?

$$\vec{v}_{\text{el}} = -b_{\text{el}} \cdot \vec{E}$$

Since the negatively charged electrons move in the opposite direction to the field direction $\vec{E}$ (and thus also in the opposite direction to the technical current direction I), the direction of the drift velocity is sufficiently described, and it is sufficient to calculate the magnitude v_{el} . With the connection

$$I = e \cdot n_{\text{el}} \cdot b_{\text{el}} \cdot E \cdot A \rightarrow E = \frac{I}{e \cdot n_{\text{el}} \cdot b_{\text{el}} \cdot A}$$

results in:

$$v_{\text{el}} = \cancel{b_{\text{el}}} \cdot \frac{I}{e \cdot n_{\text{el}} \cdot \cancel{b_{\text{el}}} \cdot A}$$

Inserting the numerical values:

$$v_{\text{el}} = \frac{8\text{A}}{1,602 \cdot 10^{-19} \text{ As} \cdot 8,5 \cdot 10^{19} \text{ mm}^{-3} \cdot 1 \text{ mm}^2}$$

$$v_{\text{el}} = 0,59 \frac{\text{mm}}{\text{s}}$$

The electrons therefore move at an average speed of 0.59 mm/s. However, since the electric field strength within the conductor travels at the speed of light, all the electrons in the conductor

start moving at practically the same time. This can be compared to a tube completely filled with marbles. The moment another marble is pushed in, a marble falls out on the other side, even though the individual marbles are moving slowly.

A Exercise tasks

A.1 Electric homogeneous field 1

A capacitor for the voltage range of 100 kV is to be realised with a dielectric in which a maximum field strength of $E = 30 \frac{kV}{cm}$ is permissible. Calculate the capacitance of the capacitor.

Task section 1:

The dielectric is first used in an (ideal) plate capacitor.

- a1) Sketch the arrangement.
- b1) Sketch the field lines in the arrangement.
- c1) Where is the location of the maximum field strength?
- d1) Sketch the course of the electric field strength from one capacitor plate to the other $|E(x)|$, parallel to the capacitor plates $|E_{Diell}(z)|$ and in the capacitor plates $|E_{Pl}(z)|$.
- e1) Draw the 50%- equipotential line.
- f1) Calculate the minimum plate spacing $d_{Pl,min}$ for the above boundary conditions.

Task section 2:

Now the dielectric between coaxial cylinders is to be inserted.

$$\left(E_r(r) = \frac{U}{r \cdot \ln\left(\frac{a}{b}\right)}; r_a : \text{Outerradius}; r_i : \text{Radius of the innercylinder} \right)$$

- a2) Sketch the arrangement.
- b2) Sketch the field lines.
- c2) Where is the location of the maximum field strength?
- d2) Sketch the course of the electric field strength $E_r(r)$ from $r = r_i$ to $r = r_a$.
- e2) Draw the 50
- f2) Calculate the minimum outer radius r_a for $r_i = 1; 2; 3; 4; 5; 10 \text{ cm}$ and the corresponding thickness of the dielectric d_i .

A.2 Electric homogeneous field 2

Between the plates of the adjacent capacitor are two insulating materials with dielectric numbers ε_{r1} and ε_{r2} . The capacitor is connected to a battery via switch S .

The following dimensions are given: Plate sub-areas $A_1 = A_2 = 100 \text{ cm}^2$; $U_B = 100 \text{ V}$; $d = 1 \text{ cm}$.

- a) The switch is closed. Let $\varepsilon_{r1} = \varepsilon_{r2} = 1$. How large are the capacitor charge Q , the electric flux density D , the electric field strength E and the capacitance C_{AB} ?

- b) The switch is closed. Let $\varepsilon_{r1} = 1, \varepsilon_{r2} = 3$. How has the total capacitance changed in general compared to point a)? The values Q_1 and Q_2 , D_1 and D_2 , and E_1 and E_2 must also be determined. What voltage U_{AB} is established at the capacitor?
- c) In the arrangement according to point a), another dielectric is introduced after the switch S is opened, so that the following applies: $\varepsilon_{r1} = 1, \varepsilon_{r2} = 3$. What voltage U_{AB} is established at the capacitor?

A.3 Electric homogeneous field 3

A capacitor for the voltage range of 100 kV is to be constructed using a dielectric in which a maximum field strength of $E = 30 \text{ kV/cm}$ is permitted.

The dielectric is first used in an (ideal) Plattenkondensator.

- a1) Sketch the arrangement.
- b1) Sketch the field lines in the arrangement.
- c1) Where is the location of maximum field strength?
- d1) Sketch the course of the electric field strength from one capacitor plate to the other $|E(x)|$, parallel to the capacitor plates $|E_{\text{Diel}}(z)|$ and in the capacitor plate $|E_{\text{Pl}}(z)|$.
- e1) Draw the 50%- equipotential line.
- f1) Calculate the minimum plate spacing $d_{\text{Pl,min}}$ for the above boundary conditions.

Now the dielectric between coaxial cylinders should be inserted.

$$\left(E_r(r) = \frac{U}{r \cdot \ln\left(\frac{a}{b}\right)}; \quad r_a : \text{Outerradius}; r_i : \text{Radius of the innercylinder} \right)$$

- a2) Sketch the arrangement.
- b2) Sketch the field lines.
- c2) Where is the location of maximum field strength?
- d2) Sketch the course of the electric field strength $E_r(r)$ from $r = r_i$ to $r = r_a$.
- e2) Draw the 50%- equipotential line.
- f2) Calculate the minimum outer radius r_a for $r_i = 1; 2; 3; 4; 5; 10 \text{ cm}$ and the corresponding thickness of the dielectric. d_i .

A.4 Charge carrier velocity

A current $I = 8 \text{ A}$ flows through a copper wire with a cross-sectional area $A = 1 \text{ mm}^2$ and a length $L = 10 \text{ m}$. One mm^3 of copper contains $8.5 \cdot 10^{19}$ atoms. It can be assumed that 1 electron per atom is involved in charge transport. ($\vartheta = 20 \text{ }^\circ\text{C}$).

- a) Determine the drift velocity of the electrons in the copper wire.
- b) What is the electric field strength E in the copper wire?

- c) How high is the voltage drop U in this copper wire?
- d) What is the resistance of the copper wire under the specified boundary conditions? What resistance does the wire assume when heated to $\vartheta_{\text{w}} = 180\text{ }^{\circ}\text{C}$? How much is the percentage increase in resistance?

B Solutions to the exercises

B.1 Electric homogeneous field 1

Hier entsteht eine Musterlösung...

B.2 Electric homogeneous field 2

Hier entsteht eine Musterlösung...

B.3 Electric homogeneous field 3

Hier entsteht eine Musterlösung...

B.4 Charge carrier velocity

Hier entsteht eine Musterlösung...